

FINETUNE UniController UC-500

Wired LAN Controller Datasheet



Ideal for:



Telecom



Hotels & Resorts



Universities



Healthcare



Public Places



Airport



Transport

UniController is an intelligent Wireless LAN Controller combined with comprehensive billing and bandwidth management solution. UniController allows network administrators centrally manage wired and wireless networks, avoid unauthorized access and track real-time activity on the network.

UniController by Indio Networks is an advanced, centralized Wireless LAN Controller designed to manage and optimize large-scale wireless networks. It enables seamless integration with Indio Access Points, offering enterprise-grade scalability, AI-powered management, and secure connectivity. UniController is ideal for airports, hotels, campuses, hospitals, and large enterprises requiring robust, policy-driven Wi-Fi control and monitoring

Key Features Overview

High Capacity & Performance

- Supports **up to 1024 Access Points** in High-Availability (HA) mode.
- Handles **up to 24,000 concurrent client devices**.
- Integrated **AI-driven Wireless and RF Optimization** for superior performance.
- Offers **Hitless Failover** and intelligent **automated load balancing** across APs.

Advanced Wireless Standards

- Fully compliant with **IEEE 802.11ax (Wi-Fi 6)** standards.
- Supports **WPA3, Enhanced Open**, and legacy Wi-Fi security protocols.

Automated Deployment & Configuration

- Enables **Zero-Touch Provisioning (ZTP)** for seamless rollout.
- Supports **Hierarchical Configuration Management** for simplified multi-site deployments.

Network Segmentation & Multi-Controller Architecture

- Supports **terminating multiple SSIDs on different controllers** using the same AP infrastructure.
- Ensures **complete network, policy, and management isolation** across segments.
- Offers **Active/Active (1:1)**, **Active/Standby (1+1)**, and **N+1 HA architectures**.

Controller Clustering & Scalability

- Scales up to **10 controllers per cluster**, ensuring high performance, redundancy, and reliability.
- Allows **seamless roaming** across Wi-Fi zones, even during controller failures or high-density events.

Client Optimization & RF Intelligence

- Mitigates sticky clients by ensuring **optimal AP association**.
- **MU-MIMO client grouping** enhances overall WLAN capacity.
- Automatically adjusts **channel and transmit power** based on live RF analytics.
- Classifies and identifies multiple **RF interference types**.
- Provides **real-time interferer charts** at AP, radio, and channel levels.
- Supports **802.11e WMM** for optimized multimedia performance.

Application Awareness & QoS

- Application-aware engine enables **per-user traffic prioritization**.
- Offers visibility and control over **1,000+ applications** through Deep Packet Inspection (DPI).
- Provides **granular user- and application-level traffic control**.
- Supports **media stream prioritization** for enhanced video QoS.
- Optimized **multicast delivery** with AP-controller session efficiency.
- Includes **IGMP snooping** to forward multicast traffic only to subscribed clients.

Security & Encryption

- Supports a comprehensive suite of encryption standards: **CCMP/AES, WEP 64/128-bit, TKIP, SSL, TLS, RC4-128, RSA-1024/2048, L2TP/IPsec, XAUTH/IPsec, PPTP**.
- **Automatically blacklists malicious clients** based on behavioral analytics.
- Supports both **IPv4 and IPv6** networks.
- Intelligent traffic control prevents non-critical traffic from impacting essential applications.

Advanced Threat Detection & Protection (WIDS/WIPS)

- Detects malicious/null SSID probe responses that target 802.11 NIC vulnerabilities.
- Prevents **AP management-frame flooding attacks** aimed at association table exhaustion.
- Blocks **broadcast disassociation** and **broadcast deauthentication** DoS attacks.
- Detects **MAC spoofing** attempts during authentication.
- Identifies **Evil Twin APs** spoofing legitimate SSIDs.
- Protects against **spoofed sleep-mode frames** used in buffer overflow attacks.
- Detects advanced wireless attack tools, including **FATA-Jack** and **Wellenreiter**.

Advanced Forensics & Spectrum Analysis

- Provides **real-time spectrum analysis** with precise interferer classification.
- Generates detailed **charts, spectrograms, heatmaps, and visual diagnostics**.
- Tracks, identifies, and locates **interferer devices** on floor maps.

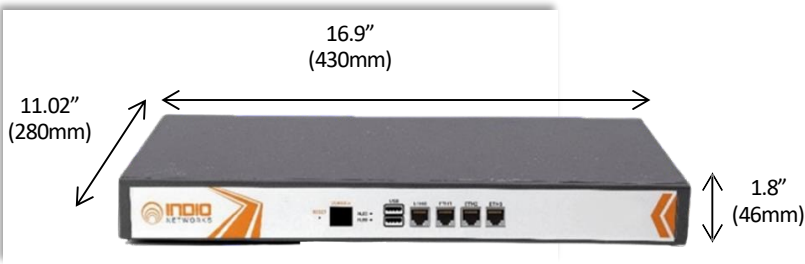
High-Performance Security & Firewall Capabilities

- Supports **2 million active firewall sessions**.
- Handles up to **24,000 concurrent IPsec sessions**.
- Delivers **40 Gbps firewall throughput**.
- Provides **20 Gbps AES-CCM encrypted throughput**.
- Features **Dynamic Segmentation** for unified wired and wireless access policy enforcement.

Next-Generation Application Visibility & Control

- Recognizes **3,000+ applications** with built-in intelligence.
- Performs deep **Layer-4 to Layer-7 traffic inspection**.
- Enables highly granular **enterprise application policy enforcement**.

Technical Specifications		Physical Property	
Interface	4 Ethernet Port (RJ-45) – 10/100/1000 base T/Tx 4 SFP+ 10G ports Intel chipset Console Port 2 USB	Dimensions	Form Factor: Standalone Physical Dimensions: 482x680x44mm Net Weight: 4kgs or 8.8lbs (approx)
Networking	Static & Dynamic routing; DNS & DHCP server; Layer 2 isolation; VLAN Tagging; SNMP v2c support; NTP (RFC 1305) support	Environmental	Operating Temperature: 5°C to 50°C Non-condensing Humidity: 8% - 90% Fanless Design
Power Rating	Input Voltage: 110-240V AC Power Consumption: 10W Heat Dissipation: 300 BTU		
Compliance	CE Emission RoHS		
System Performance	Concurrent User Sessions: 500 Data Throughput: 860Mbps (auth) 921Mbps (no-auth)		



New & Improved Features



With over two decades of experience, we have worked closely in the wireless space essentially serving over 25+ million users till date. At Indio, we have seen disruptions in the tech sphere and worked alongside them. We solve problems where we see them, we evolve, innovate and adapt. That's why, we constantly craft newer products and solutions to make your life easier. 'Simplifying connectivity' is a principle we hold dear, and we plan on bringing it to everyone, everywhere. With a strong presence in over three important geographies, being Asia, the Middle East and America, we are always available for quick and responsive customer support.

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